# SDG BLOCKCHAIN ACCELERATOR

# Mentor Feedback Report : Green Giraffe

**Program:** EMURGO : UNDP Blockchain Accelerator (Cohort 2)
**Project:** Green Giraffe : Digital Traceability & Market Access Pilot
**Country Focus:** Zambia, Angola
**Challenge Partner:** UNDP Angola : Kurima Programme
**Mentorship Focus:** Agricultural Supply Chain Traceability, Operational Deployment, Tokenized Impact Recording
**Total Mentoring Sessions:** 5
**Mentor:** Ahmed M.Amer

# 1. Initial Context and Entry Assessment

Green Giraffe entered Cohort 2 with a project that was structurally different from many of the other teams in the accelerator. Rather than building a financial product or an identity infrastructure, the team focused on a problem rooted in physical supply chains: improving transparency and market access for agricultural producers through traceability.

The project is tied closely to the Kurima Programme, which aims to support sustainable agriculture and value chain development in Angola and neighboring regions. In this context, the team proposed a blockchain-based traceability system where batches of agricultural produce could be represented digitally and tracked as they move through the supply chain.

At the start of mentoring, the team demonstrated a good understanding of the value chain problem itself. They understood that producers often face barriers when trying to sell to larger buyers or export markets because documentation, traceability, and verification processes are fragmented or unreliable.

However, the technical framing initially leaned heavily toward blockchain mechanics rather than operational realities. Early presentations focused on tokens, validators, and smart contract flows without fully mapping the day-to-day processes that farmers, aggregators, and buyers actually follow.

Mentoring therefore focused first on grounding the system in real agricultural workflows, rather than starting with technical abstractions.

# 2. Mentoring Focus: Traceability as an Operational System

One of the most important reframing moments during mentoring was recognizing that traceability systems are not primarily technical systems. They are operational systems that must integrate into existing behaviors across multiple actors in the supply chain.

In agriculture, these actors include:

- Farmers
- Aggregators
- Processors
- Transporters
- Buyers
- Certification bodies
- Government or program administrators

Each of these actors interacts with the supply chain in different ways and at different levels of digital maturity. A blockchain traceability system cannot replace these interactions; it can only support them.

The mentoring approach therefore emphasized that the system must be designed around **events in the real world**, not around token transfers.

Examples of such events include:

- Harvesting a batch of produce
- Aggregating multiple small batches
- Transporting goods to a processing facility
- Transferring ownership to a buyer
- Recording sustainability or production metrics

The blockchain layer was framed as a **recording mechanism for these events**, not the core process itself.

# 3. Mentoring Sessions and Project Evolution

## Session 1 : Mapping the Physical Supply Chain

The first session focused almost entirely on the physical supply chain.

Rather than reviewing smart contracts, the mentoring discussion asked the team to walk through a real example of how produce moves from a farm to a market buyer.

Several questions guided this discussion:

- Who records the initial batch?
- What unit of produce is being tracked?
- What happens when multiple farmers combine their produce?
- When does ownership actually change?
- Who is responsible for confirming that a transaction happened?

This exercise revealed that some of the early digital assumptions did not match how aggregation actually occurs in practice. For example, produce is often mixed and repackaged multiple times before reaching a buyer, which complicates the concept of a “single immutable batch.”

Mentor feedback encouraged the team to treat traceability as a **series of transformations**, rather than a simple chain of custody.

This insight later influenced how token transfers were modeled in the PoC.

## Session 2 : Token Design and Data Integrity

Once the operational flow was clarified, the second session moved into how that flow should be represented technically.

The team had initially proposed minting tokens that represent batches of produce. While this approach is common in traceability projects, mentoring pushed the team to think carefully about what those tokens actually represent.

A key distinction emerged between:

- **Ownership tokens** (who currently holds the produce)
- **Traceability records** (what happened to the produce)

In many systems these are conflated, but separating them improves clarity.

The discussion therefore focused on ensuring that tokens represent a **traceable unit**, while validator logic enforces the conditions under which those units can be transferred or modified.

The Aiken validators designed in the PoC enforce several important rules:

- Only authorized actors can create new batch records.
- Batch transfers must reference existing valid tokens.
- Invalid redeemers or malformed data structures are rejected.

This approach ensures that traceability records remain internally consistent even if off-chain systems behave unpredictably.

## Session 3 : Data Capture in Low-Connectivity Environments

The third session focused on a practical issue that often receives too little attention in blockchain traceability projects: how data actually enters the system.

In the contexts where Green Giraffe operates, several constraints exist:

- Farmers may have limited smartphone access.
- Connectivity can be intermittent.
- Data may initially be recorded on paper or through intermediaries.

This raises an important question: who is responsible for creating the first digital record of a batch?

The mentoring discussion emphasized that traceability systems often fail because they assume perfect digital input from the beginning. In practice, data collection must tolerate delays, manual entry, and partial verification.

The team therefore began thinking about how aggregators or program administrators might serve as the initial digital entry points, with farmers participating indirectly.

This approach aligns better with existing agricultural program workflows and reduces the burden placed directly on farmers.

## Session 4 : On-Chain vs Off-Chain Data Strategy

A recurring theme across several teams in Cohort 2 was deciding what information belongs on-chain and what does not.

For Green Giraffe, the initial instinct was to store a significant amount of traceability information on-chain. Mentoring encouraged a more restrained approach.

The discussion centered on three guiding principles:

1. The blockchain should guarantee **integrity**, not store every piece of data.
2. Sensitive or operationally heavy data should remain off-chain.
3. On-chain records should focus on events and references rather than full datasets.

Following this discussion, the team aligned on a model where:

- Key traceability events are anchored on-chain.
- Supporting documentation and metadata can be stored off-chain.
- Hash references ensure that off-chain data cannot be altered without detection.

This architecture reduces costs while preserving the core benefit of blockchain-based verification.

### Session 5 : Pilot Readiness and Program Integration

The final mentoring session focused on how the PoC could realistically support the Kurima Programme.

Unlike purely technical pilots, agricultural traceability pilots must integrate with ongoing development programs, farmer networks, and government stakeholders.

This session reviewed:

- How tokens would be created in real pilot conditions
- Who would operate the system day to day
- What stakeholders would need to see to trust the system

One important outcome was recognizing that the system should initially focus on **demonstrating traceability events**, rather than attempting to capture the entire supply chain immediately.

Starting with a narrower scope;such as tracking specific high-value product batches;would allow the team to demonstrate value without overwhelming operational partners.

## 4. Overall Mentor Assessment

Green Giraffe showed consistent improvement throughout the accelerator. The team entered the program with a technically interesting concept but gradually shifted toward a more realistic and operationally grounded system.

The most important progress made during mentoring was the move away from a technology-first narrative toward a **supply-chain-first approach**.

## 5. Strengths Observed

Several strengths stood out during the mentoring process.

First, the team showed a strong willingness to adapt its design when operational realities were discussed. Rather than insisting on an idealized digital model, the team gradually incorporated feedback about agricultural workflows.

Second, the technical implementation using Aiken validators was solid. The contracts enforce authorization rules and reject invalid transactions, which is essential for maintaining data integrity in a traceability system.

Third, the project is well aligned with the goals of the Kurima Programme and broader SDG objectives related to sustainable agriculture and responsible production.

# 6. Challenges and Risks

Despite the progress made, several challenges remain.

The most significant risk is not technical but operational. Traceability systems rely heavily on consistent data input, and ensuring that supply chain actors record events reliably will require strong operational support.

Another challenge is managing expectations. Blockchain-based traceability can improve transparency, but it cannot automatically guarantee the accuracy of all real-world data inputs.

Finally, the system will need to demonstrate clear value to supply chain participants. If recording traceability events adds too much friction without clear benefits, adoption may stall.

# 7. Mentor Conclusion

Green Giraffe’s work during Cohort 2 demonstrates the potential for blockchain technology to support more transparent agricultural supply chains. The project made meaningful progress in grounding its technical design in real operational processes.

The PoC provides a credible foundation for pilot testing within the Kurima Programme, particularly if the initial scope remains focused and manageable.

Going forward, success will depend less on new technical features and more on operational execution: ensuring that the system integrates smoothly into existing agricultural workflows and provides clear value to the participants involved.